1 Write $\frac{2}{5}$ as a decimal.

0.4

(Total for Question 1 is 1 mark)

---

2 Write 0.03 as a fraction.

$\frac{3}{100}$

(Total for Question 2 is 1 mark)

---

3 5 calculators cost £31.25 $5 \times 6 = 30$
3 pens cost £2.91 $3 \times 10 = 30$

1 mark for correct method

Work out the cost of 30 calculators and 30 pens.

$31.25 \times 6 + 2.91 \times 10$

$= £216.60$

£ 216.60

(Total for Question 3 is 2 marks)

---

4 A square has a perimeter of 32 cm.

Find the area of the square.

$\frac{32}{4} = 8$ each side is 8cm

1 mark for 8 (cm)

$8 \times 8 = 64 \text{ cm}^2$

64 cm^2

(Total for Question 4 is 2 marks)

5 Here are six cards. Each card has a number on it.

15	14	9	12	16	21

(a) Work out the range of the numbers on the cards.

$21 - 9$

12

(1)

(b) Work out the mean of the numbers on the cards.

$$\frac{15 + 14 + 9 + 12 + 16 + 21}{6} = 14.5$$

1 mark for correct method

14.5

(2)

(Total for Question 5 is 3 marks)

---

6 Work out $\dfrac{3.15 + 2.8^2}{2.06}$

1 mark for 10.99

Write down all the figures on your calculator display.

5.334951456

(Total for Question 6 is 2 marks)

---

7 $L = 9m + 2n$

Work out the value of L when $m = -3$ and $n = 4$

$9(-3) + 2(4) = -19$

1 mark for correct substitution

-19

(Total for Question 7 is 2 marks)

8 (a) Solve $3x = 36$

$x =$ 12

(1)

(b) Solve $\frac{24}{y} = 6$

$y =$ 4

(1)

(c) Solve $a - 8 = 17$

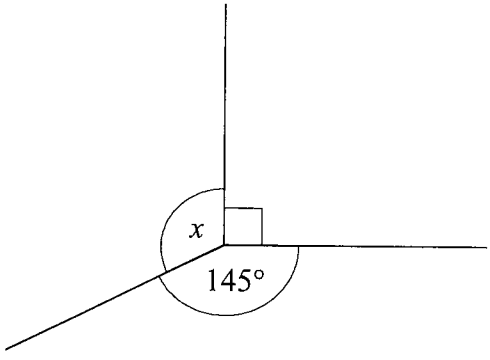
$a =$ 25

(1)

(Total for Question 8 is 3 marks)

---

9 Work out the size of the angle marked x.



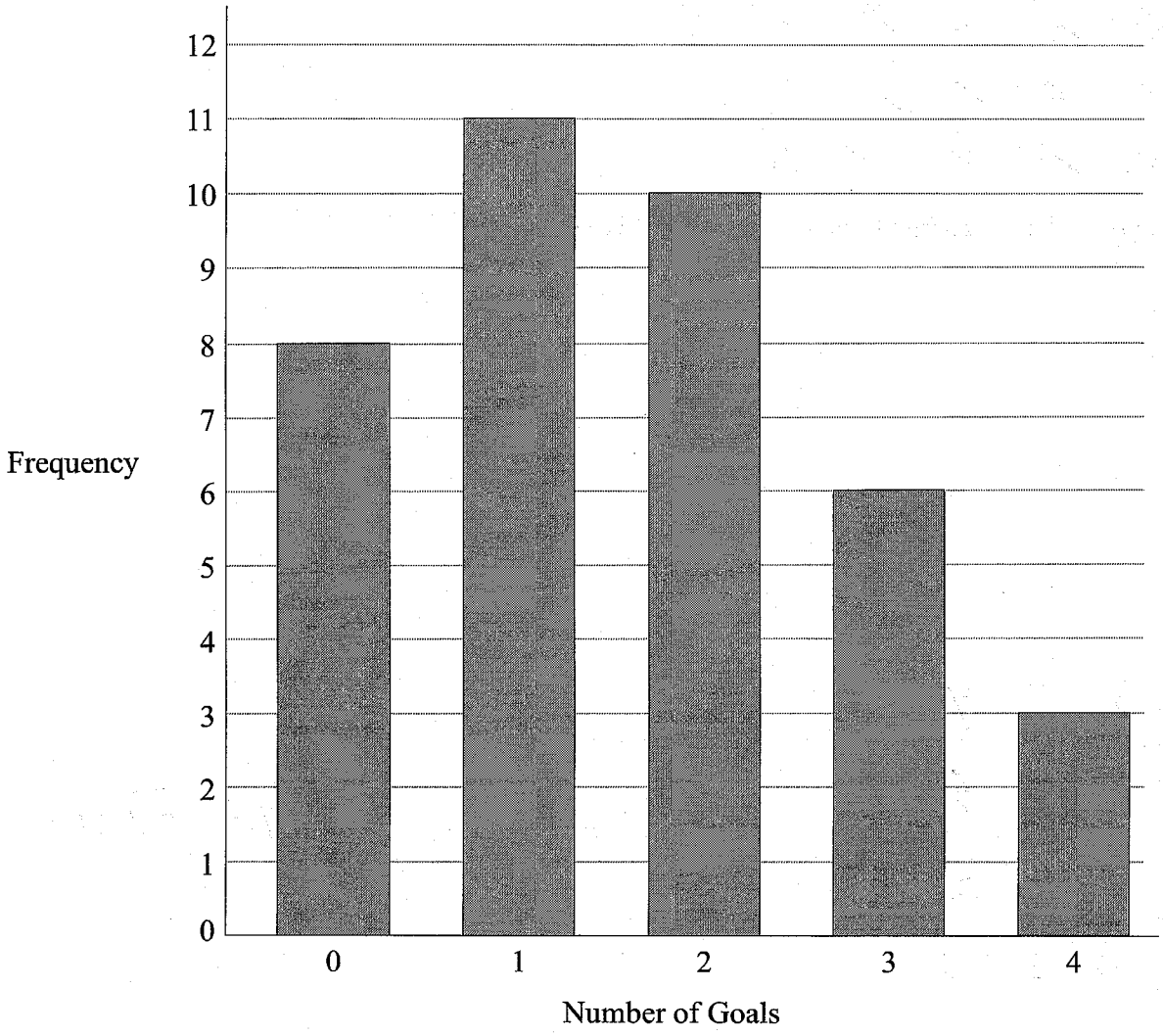
360 − 90 − 145

1 mark for correct method

125 °

(Total for Question 9 is 2 marks)

10 Here is a bar chart showing the number of goals scored in a game by a football team in a season.



(a) What is the modal number of goals scored.

1

(1)

(b) Work out the total number of goals were scored in total by the team.

1 mark for (8), 11, 10, 6 and 3

$11 \times 1 = 11$

$10 \times 2 = 20$

$6 \times 3 = 18$

$3 \times 4 = 12$

$11 + 20 + 18 + 12 = 61$

61

(3)

1 mark for correct method

(Total for Question 10 is 4 marks)

11 Here are the heights, in cm, of 15 plants.

50	39	52	53	25
32	29	45	35	33
29	27	46	44	41

Draw an ordered stem and leaf diagram to show this information.

1 mark for just one mistake

2	5 7 9 9
3	2 3 5 9
4	1 4 5 6
5	0 2 3

Key: 2/5 = 25cm

1 mark for correct key

(Total for Question 11 is 3 marks)